

CS215 Assignment-1

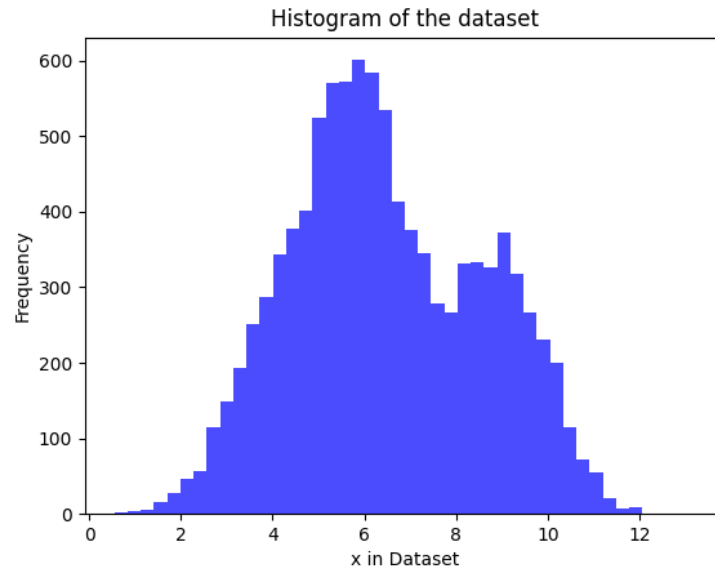
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1. Fitting Data Task B

Solution:

Graphically the the normal distribution parameter μ seems to be around 6.



Histogram plot

Task C

Solution:

Let $X \sim \text{Bin}(n, p)$

First moment of Binomial Distribution, μ_1^{bin}

$$\mu_1^{bin} = E(X) = \sum_{x=0}^n x \binom{n}{x} p^x (1-p)^{n-x}$$

$$\mu_1^{bin} = np \sum_{x=1}^n \binom{n-1}{x-1} p^{x-1} (1-p)^{n-x}$$

$$\mu_1^{bin} = np(1 - p + p)^{n-1}$$

$$\mu_1^{bin} = np$$

Second moment of Binomial Distribution, μ_2^{bin}

$$\mu_2^{bin} = E(X^2) = \sum_{x=0}^n x^2 \binom{n}{x} p^x (1-p)^{n-x}$$

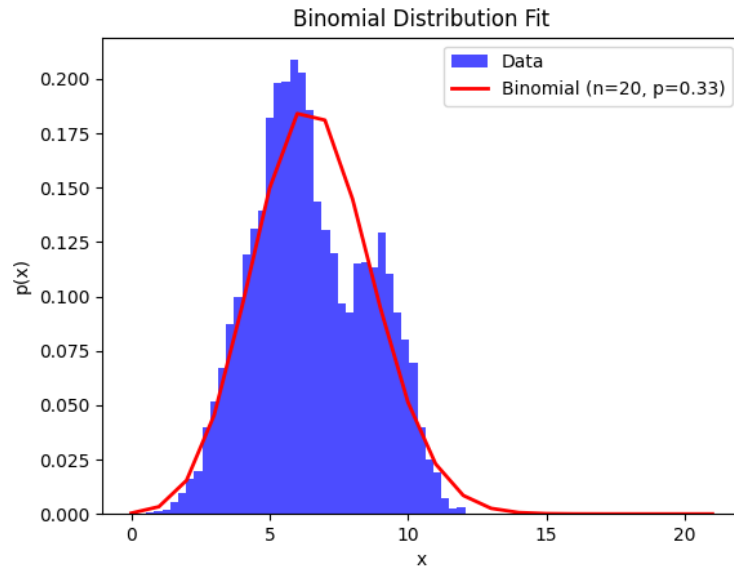
$$\mu_2^{bin} = \sum_{x=0}^n \{x(x-1) + x\} \binom{n}{x} \binom{n-1}{x-1} \binom{n-2}{x-2} p^x (1-p)^{n-x}$$

$$\mu_2^{bin} = n(n-1)p^2 \left\{ \sum_{x=2}^n \binom{n-2}{x-2} p^{x-2} (1-p)^{n-x} \right\} + np$$

$$\mu_2^{bin} = n(n-1)p^2(1-p+p)^{n-2} + np$$

$$\mu_2^{bin} = n(n-1)p^2 + np$$

$$\mu_2^{bin} = np(1-p) + n^2p^2$$



Binomial Fitting

Task D

Solution:

Lets calculate i^{th} moment of gamma function given by:

$$f(x, \alpha, \theta) = \frac{x^{\alpha-1}}{\Gamma(\alpha)\theta^\alpha} e^{-\frac{x}{\theta}}$$

$$\begin{aligned}
E[X^i] &= \int_0^{\infty} \frac{x^k}{\Gamma(\alpha)\theta^\alpha} x^{\alpha-1} e^{-\frac{x}{\theta}} dx = \frac{1}{\Gamma(\alpha)\theta^\alpha} \int_0^{\infty} x^{(\alpha+i)-1} e^{-\frac{x}{\theta}} dx \\
&= \frac{\Gamma(\alpha+i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} \cdot \int_0^{\infty} \frac{1}{\Gamma(\alpha+i)\theta^{\alpha+i}} x^{\alpha+i-1} e^{-\frac{x}{\theta}} dx \\
&= \frac{\Gamma(\alpha+i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} \cdot (\text{p.d.f. of } \Gamma(\alpha+i, \theta) \text{ integrates to } 1) \\
&= \frac{\Gamma(\alpha+i)\theta^{\alpha+i}}{\Gamma(\alpha)\theta^\alpha} = \frac{\theta^i \Gamma(\alpha+i)}{\Gamma(\alpha)} \\
&= \frac{\theta^i (\alpha+i-1)(\alpha+i-2) \cdots \alpha \Gamma(\alpha)}{\Gamma(\alpha)} = \theta^i (\alpha+i-1) \cdots \alpha.
\end{aligned}$$

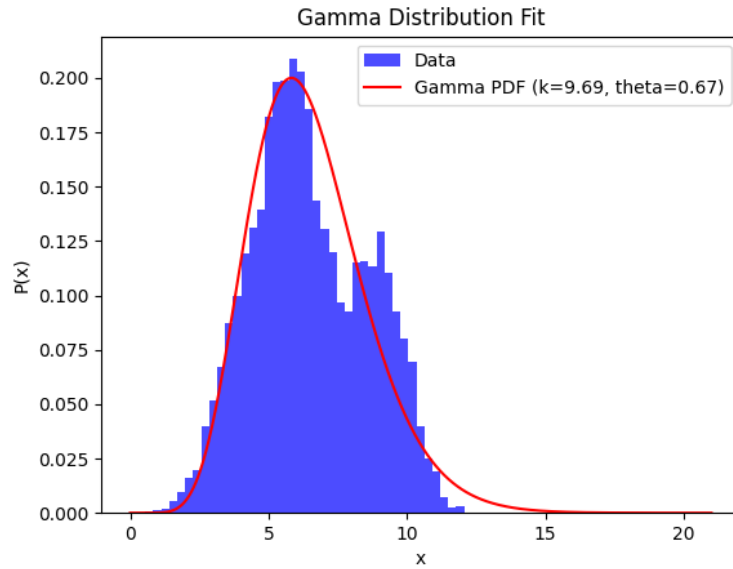
Thus,

First moment μ_1^{gamma} is given by (replacing α by k in above derivation),

$$\mu_1^{gamma} = k\theta$$

And second moment μ_2^{gamma} is given by,

$$\mu_2^{gamma} = k(k+1)\theta^2$$



Gamma Fitting

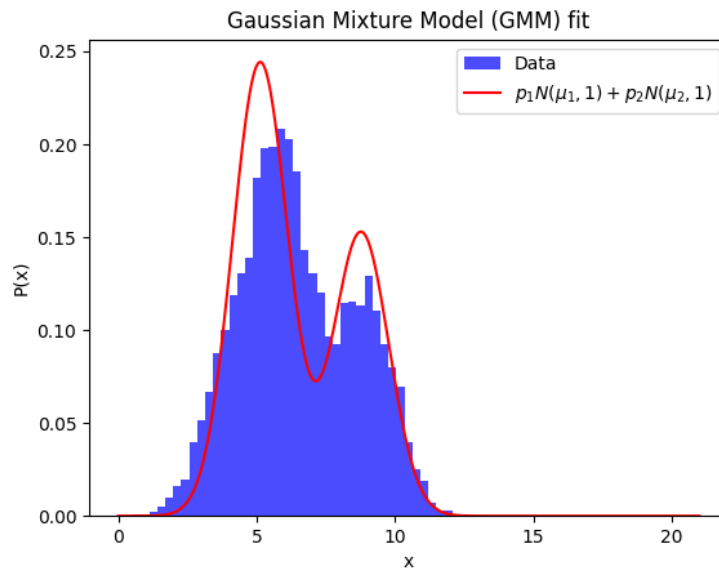
Task E

Solution:

The likelihood of binomial (-2.1570681154346776) is a bit greater (closer to zero) than likelihood of gamma distribution (-2.1608217722067953) indicating that binomial is a better fit.

Task F**Solution:**

The likelihood of Mix Guassian distribution obtained is -2.183038744911319. This however is not better than any of the previous two!!



GMM Fitting

Ungraded Bonus Question**Solution:**

In order to get a better fit, lets choose again a mixture of guassian distribution. The difference from previous case is that instead of two guassian, lets take three distribution superimposed over each other with probabilities p1, p2 and p3, mean as μ_1 , μ_2 and μ_3 and variance of all of them as 1.

The two new moments (fifth and sixth) can be derived giving the following expression:

$$\begin{aligned} \text{moment5}^{th} &= \mu^5 + 10\mu^3\sigma^2 + 15\mu\sigma^4 \\ \text{moment6}^{th} &= \mu^6 + 15\mu^4\sigma^2 + 45\mu^2\sigma^4 + 15\sigma^6 \end{aligned}$$

Similar to the previous method, calculating 5th and 6th moment of given dataset and solving the equations using fsolve fuction of scipy, we were getting impossible soltions as the jacobian failed to converge.

Thus making an initial guess and hypertuning the probabilities of the GMM, we finally arrived at a better fit whose GMM is as follows:

$$Y \sim p1 \cdot N_1(\mu1, 1) + p2 \cdot N_2(\mu2, 1) + p3 \cdot N_3(\mu3, 1)$$

$$where, p1 = 0.5018740341610936$$

$$\mu1 = 6.02960769428528$$

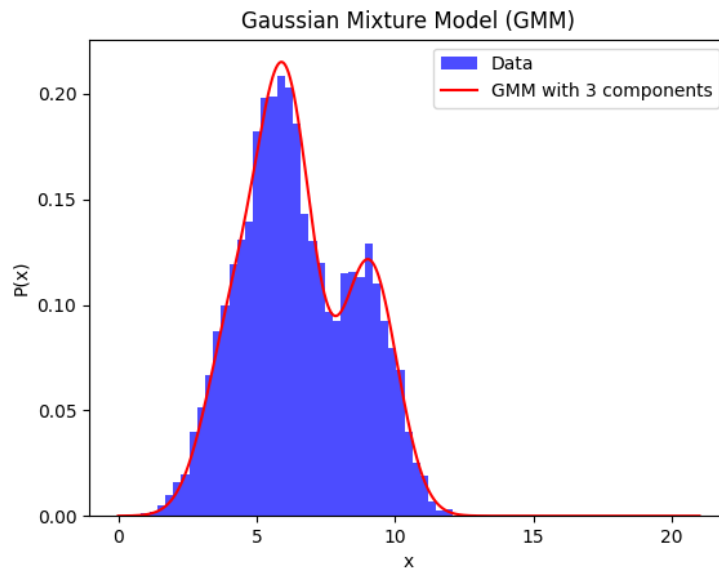
$$p2 = 0.2996456511927248$$

$$\mu2 = 9.07436305441652$$

$$p3 = 0.200116472923$$

$$\mu3 = 4.1000265473$$

Thus the final fit is:



Final GMM Fitting